

MSRA: A Parameter-Efficient Spectral Correction for Local Attention — Where It Helps, and Where It Fails

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Abstract

Restricting attention to a sliding window makes long documents affordable, but it discards everything beyond the window. This paper asks what can be recovered when the model is already trained, its weights are frozen, and the budget for new parameters is only a few tens of thousands. We propose Multi-scale Spectral Residual Attention (MSRA): a far-field branch grafted onto the last few layers of a frozen model, which condenses out-of-window history through smoothly distance-decayed weights and adds it back to the attention output—roughly 9K–45K trainable parameters, plus minimal 8-to-16-scalar probes for attribution. On PG-19 long-form fiction, the branch closes up to 31.7% of the perplexity gap between local and full attention, with gains stable across two architectures, multiple seeds, and the official test partition. The attribution proves sharper than the headline: 16 learnable output-gain scalars alone close 26.4%, returns accelerate with grafting depth, the carefully designed kernel structure and gating contribute only secondarily—and a budget-matched LoRA closes 87.4%, indicating that the gap is mostly generic capacity rather than a missing far-field pathway. Retrieval experiments complete the map: the correction is a statistical smoother that helps just beyond the window, slightly hurts precise in-window prediction, recovers essentially nothing at genuine long distance, and does not transfer to task metrics—while slowing prefill two-to-three-fold and leaving decoding unaffected. These results bound what a lightweight spectral correction can and cannot recover, forming a reproducible decision reference for local long-context design.

1 Introduction

One of the mainstream ways modern language models handle long sequences (Vaswani et al., 2017)

is to let each token attend only to the most recent w tokens. This design compresses the cost of attention from quadratic to linear in sequence length, making documents of thousands of words practical to process. Yet anyone who has read a novel knows how much a hint planted in chapter three can matter—and to a windowed model, chapter three might as well not exist.

A large body of work tries to close this gap (see Tay et al., 2022, for a survey): sparse attention lets selected positions see farther (Beltagy et al., 2020; Zaheer et al., 2020; Child et al., 2019; Roy et al., 2021), linear attention replaces the standard mechanism with kernel approximations (Katharopoulos et al., 2021; Wang et al., 2020), sparse and memory-efficient variants continue to emerge (Kitaev et al., 2020; Dao et al., 2022), state-space models take a different route to memory altogether (Gu and Dao, 2023; Peng et al., 2023), and hybrid local–global architectures add a global pathway alongside the window—a lineage reaching back to segment-level recurrence (Dai et al., 2019) and compressive memory (Rae et al., 2020), most recently in hybrid local–global designs (Arora et al., 2024a; Munkhdalai et al., 2024). Linear position biases extend windows at inference time (Press et al., 2023), and streaming designs bound the cache outright (Xiao et al., 2024). Most of these solutions replace the attention mechanism itself, or require retraining the model end to end. Some settings afford no such luxury—the model is already deployed, its original weights are untouchable, and the budget for new parameters is a few tens of thousands. Our question follows: under constraints this tight, how much of the lost information can a spectral far-field correction actually recover? And what is the recovered portion made of?

We propose MSRA: grafting a frequency-domain residual branch onto the last L layers of a frozen model, training only that branch. The setting has clear practical relevance—deployed

*Contact author. Code and data: <https://github.com/TianZH-XJU/MSRA-EMNLP-GLM2026>

models that cannot be retrained, extremely low adaptation budgets — and it doubles as a clean attribution workbench, because every component of the branch (output gain, kernel structure, multi-scale, gating, grafting depth) can be switched on or off independently and examined one by one.

With this workbench we measured thirteen configurations systematically, and the picture that emerged is sharper than expected. The perplexity improvement is real and stable across seeds, yet its source is not the spectral structure we designed but two plainer factors: a per-layer learnable output strength, and the depth over which the branch is spread. Spending the same parameter budget on a generic low-rank adapter yields nearly three times the recovery. And on tasks that require retrieving specific distant evidence, the correction has nothing to offer — what it can do is calibrate the model’s expectations near the window’s edge, not fetch facts from afar. On efficiency, the default configuration slows the reading-in stage by two to three times, while word-by-word generation runs at exactly the speed of the uncorrected model.

The contribution of this paper is therefore not another efficient attention variant, but a boundary map: what a minimally-budgeted spectral correction recovers, what it cannot, and where its costs lie. Every number is anchored to one reproducible evaluation protocol; for practitioners designing local long-context models, the map serves directly as a decision reference.

2 Related Work

Sliding-window and sparse attention. The idea of sliding-window attention builds on a line of structured sparse attention (Beltagy et al., 2020; Zaheer et al., 2020; Child et al., 2019). Longformer and BigBird let selected positions see farther, keeping attention affordable; Mistral showed that a sliding window suffices as a production default (Jiang et al., 2023). These methods bake the sparse structure into the model from the start of training, letting it grow together with the weights; we study the complementary question of what can still be retrofitted once the model is fixed.

Kernelized attention. Linear attention and Performer rewrite query–key similarity into a form that can be aggregated in advance, sidestepping quadratic cost (Katharopoulos et al., 2021; Choromanski et al., 2021; Wang et al., 2020); long-convolution models compute the same aggregation

in the frequency domain (Poli et al., 2023). The Hedgehog analysis (Zhang et al., 2024) is especially relevant to us: it shows that positive feature maps of the ELU+1 family inherently lack the sharpness of softmax attention and struggle to single out one exact position. Our far branch uses precisely such a map, and our empirical findings — statistical smoothing works, precise retrieval does not — provide direct evidence for that theoretical expectation, echoing the recall-throughput tradeoff measured for efficient architectures (Arora et al., 2024b) and the copying gap between attention and fixed-state models (Jelassi et al., 2024).

Hybrid local–global architectures. BASED couples exact local attention with a global linear branch (Arora et al., 2024a); Infini-attention extends attention with compressive memory (Munkhdalai et al., 2024). Both train the global pathway end to end; MSRA instead freezes the base and grafts a minimal residual, answering a diagnostic question: to what extent can a global pathway be retrofitted after the fact? Our answer — yes for overall language-modeling quality, no for precise retrieval — offers a reference point for such designs.

Parameter-efficient adaptation. The parameter-efficient adaptation literature has long shown that tens of thousands of parameters can substantially change a frozen model’s behavior — from adapter layers (Houlsby et al., 2019) and continuous prompts (Li and Liang, 2021) to LoRA (Hu et al., 2022) and long-context fine-tuning (Chen et al., 2024). We compare the spectral correction against same-budget LoRA, and against a plain depthwise convolution, under identical data, training, and evaluation protocols. To our knowledge, such same-protocol comparisons of “specialized structure versus generic adapter” are rare in the long-context correction literature — and it is exactly this comparison that yields one of this paper’s weightiest conclusions.

3 Method: Multi-scale Spectral Residual Attention

3.1 Intuition

Picture each token as a listener in a meeting room. It keeps a notebook with detailed records of the most recent statements — that is the exact attention within the window; of everything said earlier, only a vague impression remains. MSRA’s idea is to

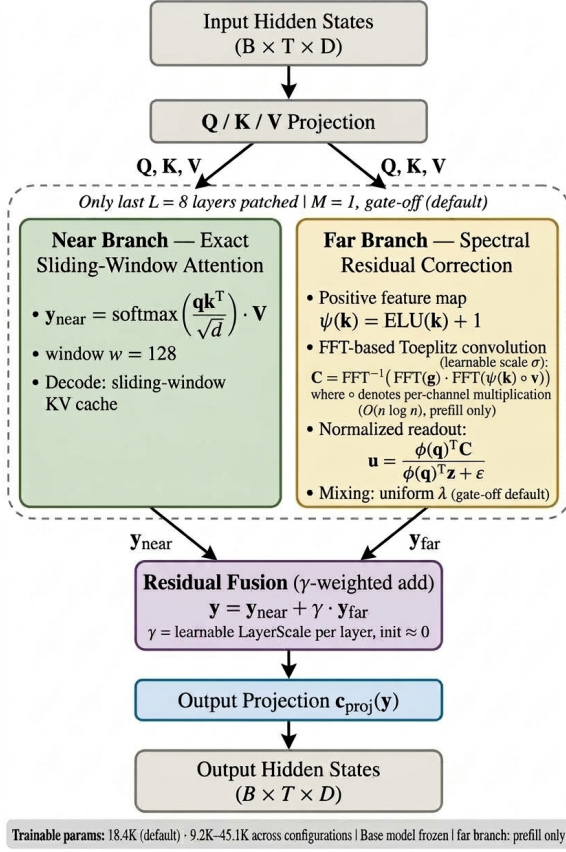


Figure 1: MSRA architecture. The last L layers of a frozen model receive a far-field residual alongside exact sliding-window attention: an exponentially decayed, FFT-computed aggregate of out-of-window key–value pairs, read through positive feature maps and scaled by a per-layer learnable gain γ .

hand the listeners in the back row a pair of binoculars: everything beyond the window is weighted by distance, farther statements weighted less, and condensed into a supplementary summary tucked into the notebook (Figure 1). The binoculars themselves are extremely simple — about twenty-three hundred parts per layer — and the listeners’ original note-taking is left untouched.

3.2 Definition

The base model is entirely frozen. On the attention output of the last L layers, a far-field residual is added to the exact local result:

$$y_i = y_i^{\text{near}} + \gamma \cdot y_i^{\text{far}}. \quad (1)$$

Here y_i^{near} is the original output of sliding-window attention, y_i^{far} is the output of the far branch, and γ is a per-layer learnable gain — the LayerScale of Touvron et al. (2021) — initialized near zero, so the model starts smoothly from a purely local state.

For each expert m — think of it as one “field of view” — the far branch computes two weighted sums:

$$C_i^{(m)} = \sum_{j \in F(i)} g_m(i-j) \psi_m(k_j) v_j^\top, \\ z_i^{(m)} = \sum_{j \in F(i)} g_m(i-j) \psi_m(k_j), \quad (2)$$

$$u_i^{(m)} = \frac{\varphi(q_i)^\top C_i^{(m)}}{\varphi(q_i)^\top z_i^{(m)} + \varepsilon}, \quad y_i^{\text{far}} = \sum_m \lambda_{im} u_i^{(m)}. \quad (3)$$

$F(i)$ is the far-field evidence set of position i — every position beyond the window, in the past direction; $g_m(\Delta) = \exp(-\Delta/\sigma_m)$ is an exponential decay kernel whose falloff speed is set by the scale σ_m ; φ and ψ are ELU+1 positive feature maps (Clevert et al., 2016), which keep the denominator positive and training stable; λ are the mixing weights across fields of view, produced either by a per-token gating network in the mixture-of-experts tradition (Shazeer et al., 2017) (gate-on) or by simple uniform averaging (gate-off). In plain terms, these equations gather out-of-window history with distance-decayed weights, then let the current token’s query read from that aggregate in a gentle, normalized way.

3.3 Design Conventions

To keep attribution clean, we compress the movable parts to a minimum. The kernel shape is fixed as exponential decay, with only its scale $\sigma = \exp(\log \sigma)$ learnable; the base weights are all frozen; trainable parameters live only inside the correction branch — the feature maps, the kernel scales, the optional gate, and the per-layer output gain γ . With M experts, each carries its own feature map and scale, initialized at log-uniform spacings (decay lengths of roughly 20 to 400 tokens), forming a set of multi-scale views from near to far.

3.4 Efficient Implementation

Because the exponential kernel depends only on distance, both sums above are convolutions and can be computed in $O(n \log n)$ with the fast Fourier transform; at inference the transformed kernels are cached by scale rather than recomputed.

A model works in two stages. The reading-in stage (prefill) processes the whole prompt at once, and the far branch computes only there; afterwards, the word-by-word generation stage (decoding) falls back to pure local attention through a

sliding-window KV cache — picture the cache as a notepad that grows as text is generated, with each new word consulting only the most recent $w+1$ entries instead of rereading the document. The far branch is structurally unable to fire during decoding, so generation runs at exactly the speed of uncorrected local attention (Table 5).

3.5 Default Configuration

Unless stated otherwise, the default configuration is single-scale ($M=1$, no gating), grafted on the last 8 layers, window $w=128$, feature rank $r=16$: 2,306 trainable parameters per layer, 18,448 in total on LLaMA 3.2-1B (Grattafiori et al., 2024). GPT2-XL experiments (Radford et al., 2019) use the same configuration. All linear projections in the branch are zero-initialized — a ReZero-style residual initialization (Bachlechner et al., 2020) — so the starting point of training is exactly the uncorrected model.

4 Experimental Design

4.1 Data and Evaluation Protocol

Language modeling, and why “gap closure.” Language-modeling experiments run on PG-19 long-form fiction (Rae et al., 2020), with documents split 75/25 for training. The evaluation set is a one-time fixed list of 16 text chunks drawn from books disjoint from the 64-book training pool; the sampling rule, the selected books, and per-book token counts are recorded in a manifest file with a SHA-256 checksum — every configuration answers the same exam paper, which is what makes the comparisons meaningful. All configurations are additionally verified on the official PG-19 test partition, which training never touches, with consistent conclusions.

The primary metric is *gap closed*. The perplexity difference between local and full attention is the price of everything the window throws away; the fraction of that difference a correction recovers is its score. Concretely, for LLaMA at 512 tokens: local attention sits at 371.6, full attention at 17.0, and MSRA at 327.1 — a closure of $(371.6 - 327.1)/(371.6 - 17.0) \approx 12.5\%$. We report gap closure throughout and flag the raw-reduction caliber explicitly wherever it is quoted. Perplexity itself can be read as the model’s average “surprise” at the next word; lower is better.

Retrieval: needle-in-a-haystack (NIAH). Does perplexity improvement mean the model can ac-

tually *find* distant information? The most direct exam is the needle-in-a-haystack protocol popularized by Kamradt (2023): hide a synthetic sentence (“The secret key is XYZZY-789”) inside pages of natural fiction, ask for it at the very end of the sequence, and measure the negative log-likelihood on the *answer tokens* — the answer sits in plain sight somewhere in the text, and we measure how confident the model is when predicting it. Examples are bucketed by the distance from evidence to question: near (≤ 128 , inside the window), mid (129–256, just outside), and far (> 256). Two protocol rules are enforced at runtime: insertion happens before any truncation, and every needle is asserted to survive truncation intact (the aggregate survival rate is 1.0); the loss falls strictly on answer tokens. Two disclosures are made plainly: at the smallest intended distances the needle shifts slightly leftward to fit, so its actual distance exceeds the nominal one while remaining inside the window; and estimating the query position introduces a boundary slide of about six tokens between buckets.

Task metrics (LongBench). Real-world capability is checked with official LongBench metrics (Bai et al., 2024) (F1 for question answering, ROUGE-L for summarization) on seven English tasks, under greedy decoding, with the three compared models sharing identical prompts (left-truncated to 2,048 tokens) and generation settings. A needle-aware RULER retest (Hsieh et al., 2024) serves as a complementary control.

Efficiency. Prefill throughput (512–4,096 tokens) and decode throughput are measured pairwise on the same machine within the same session, anchored to a freshly measured local baseline.

Training protocol. The training protocol is uniform across configurations: AdamW (Loshchilov and Hutter, 2019) (learning rate 2×10^{-4} for spectral parameters; the adapter baselines follow their convention at the more conservative 10^{-4}), cosine annealing, 3,000 steps, document-level sampling. Every trained structure runs at least 2 random seeds, primary configurations 3 to 5; the gain-only probes run 1 seed.

4.2 What We Compare Against

Judging a correction branch requires a ladder of controls. The two outermost goalposts are full attention and local attention: the former sees the whole text and defines the bottom of the gap; the

Table 1: Compared configurations and trainable parameter counts. Every multi-scale row carries a gate-off control; the gain-only probes train one scalar per patched layer.

Category	Configuration	Params
Baselines	Full / Local att. (w=128)	0
Spectral	Gain-only probes (L8/L16)	8–16
	M1L4 / M1L8 / M1L16	9.2K / 18.4K / 36.9K
	M4L4 / M4L8 (gate-off ctl.)	21.5–45.1K
	rank-1 LoRA @ o_proj	32.8K
Generic adapters	Causal dw. sep. conv (k=8)	131.1K
Placebo	Untrained random kernel	0

Table 2: PG-19 perplexity on the fixed 16-chunk evaluation set. Gap closed is shown per row; seed counts are 5/3/1 across rows, every seed positive. Gains relative to local span 10.4–13.1% at the configuration level; replicated on the official PG-19 test partition (+10.1%).

Model	Length	Full	Local	MSRA	Gap cl.
GPT2-XL	512	42.6	2,649 $\pm$ 571	2,373 $\pm$ 535	10.6%
LLaMA	512	17.0	371.6	327.1	12.5%
LLaMA	1024	18.1	545.7	474.3	13.5%

latter is the uncorrected base and defines the starting line. Within the spectral family, the ladder runs from simple to elaborate: at the bottom sit the gain-only probes, which train nothing but the per-layer output-strength scalars (8 scalars for the 8-layer skeleton, 16 for the 16-layer one) with everything else frozen; above them are the single-scale family at three depths (M1L4, M1L8, M1L16) and the multi-scale family (M4L4, M4L8), each multi-scale row paired with a gate-off control. Outside the spectral family stand two generic opponents: a budget-matched rank-1 LoRA adapter (Hu et al., 2022) and a plain causal depthwise convolution — neither contains any spectral structure, and they answer the question “what happens with the same budget but no structural priors?” Last comes a placebo control: a randomly initialized, never-trained branch, verifying that the gains come from learning rather than from injection itself. All configurations and their parameter counts appear in Table 1.

5 Results

5.1 Language Modeling: Gains Are Real and Stable

Three facts deserve attention first (Table 2). The degradation of local attention on fiction is severe — LLaMA’s perplexity at 512 tokens jumps from 17 to 372, not a loss of a few percent but a more-than-twenty-fold worsening, which says the infor-

Table 3: Attribution ladder. All values are independent-process evaluations on the same fixed evaluation set; seed counts are 1–3 per row (supplementary material).

Correction	Params	Gap closed
Untrained random kernel	18.4K	0.0%
Gain-only (M1L8)	8	10.0%
Gain-only (M4L8)	8	10.1%
Gain-only (M1L16)	16	26.4%
M1L4 (gate-off)	9.2K	5.3–5.4%
M4L4 (gate-off)	21.5K	6.1%
M1L8 (default)	18.4K	12.5%
M4L4 (gate-on)	22.5K	8.6–8.7%
M4L8 (gate-off)	43.0K	15.4%
M4L8 (gate-on)	45.1K	25.7%
M1L16 (gate-off)	36.9K	31.7%
Depthwise sep. conv (k=8)	131.1K	78.3%
rank-1 LoRA	32.8K	87.4%

mation outside the window is nearly indispensable for long novels. The correction’s effect is stably positive: configuration-mean gains span 10.4% to 13.1% (10.4% for GPT2-XL, 12.0% for LLaMA at 512), and every individual seed is positive — GPT2-XL’s five per-seed gains range from 9.8% to 10.9%, none negative. Replacing the evaluation set wholesale with the official test partition leaves the conclusion standing (+10.1% on LLaMA), ruling out a lucky choice of exam paper.

In budget terms: 18,448 parameters — about fifteen millionths of the base (0.0015%) — buy roughly a 12% perplexity reduction. Whether that trade is worth making depends on what the 12% is made of, which is exactly what the next two sections take apart.

5.2 The Attribution Ladder: Where the Gains Come From

Read Table 3 as a sequence of single moves: each row, relative to the ones above, trains one more component or deepens the graft. The placebo control offers an early reassurance — a randomly initialized branch closes 0.0%, so everything above it is learned.

Finding 1: the bulk of the gain comes from an unexpectedly humble component. Training only the per-layer output-gain scalars — 8 of them on the M1L8 skeleton — already closes 10.0%; switching to a multi-scale skeleton while still training only the 8 scalars gives an essentially identical 10.1%. At sixteen layers, 16 scalars alone close 26.4%. At both depths the pure gain channel accounts for 80–83% of the full structure’s recovery, with the complete spectral apparatus — feature

maps, kernel scales, gating — adding only 2.5 to 5.3 points on top. More intriguing is how the gain channel itself scales: doubling depth lifts closure from 10.0% to 26.4%, far more than linear. Section 5.3 shows what this channel really is: a calibration of the output.

Finding 2: depth is the dominant lever. The single-scale family rises monotonically with grafting depth and steepens as it goes: 5.4% at four layers, 12.5% at eight, 31.7% at sixteen. Stated in budget-matched form the conclusion turns sharper still — redirecting the M4L8 budget (45.1K parameters, 25.7%) into depth yields M1L16, which uses fewer parameters (36.9K) and closes more (31.7%); redirecting the M4L4 gate-on budget (22.5K, 8.6%) into depth yields M1L8, again fewer parameters (18.4K) and more closure (12.5%). An honest boundary needs stating: at four layers the raw cross-cell comparison runs the other way (8.6% for the gated multi-scale cell against 5.4% for single-scale), which is exactly why we state the claim in budget-matched rather than point-wise form.

Finding 3: multi-scale and gating are genuine but secondary. Without gating, multi-scale adds 0.7 and 2.9 points at depths four and eight; the gate’s contribution grows with depth, from 2.5 points at four layers to 10.3 at eight. The trained gate does learn selectivity — its average entropy is only 62.8% of uniform, and perturbing the query token moves the gate 113 times as strongly as perturbing far-context tokens. Kernel scales in the multi-scale configurations drift modestly from initialization (−10% to +14%), suggesting the initial spacing was largely sound and training plays a refining role; in the single-scale configurations the trajectories beyond the gain are smaller still (supplementary material).

Finding 4: budget-matched generic adapters dominate. A budget-matched rank-1 LoRA — a generic adapter with no spectral structure, simply a low-rank pair alongside the attention output projection — closes 87.4% with 32.8K parameters, 2.8 times the spectral family’s best; even a plain depth-wise convolution, given seven times the budget, reaches 78.3%. Flanked by two generic adapters, the conclusion is hard to escape: the gap between local and full attention is mostly a *generic capacity deficit*, not a far-field information gap that only spectral structure can express — the spectral form is not a necessary inductive bias for this task.

Table 4: NIAH answer-token NLL ($w=128$, 2,048 tokens; 33 needles, 11 per bucket; survival asserted per example, aggregate rate 1.0). Parentheses give closure of the local–full NLL gap — negative means the correction is worse than local. $\pm$ is the across-example standard deviation.

Configuration	Near	Mid	Far
Full	0.82 $\pm$ 0.15	0.80 $\pm$ 0.17	0.91 $\pm$ 0.11
Local	2.37 $\pm$ 0.83	9.35 $\pm$ 1.79	9.75 $\pm$ 0.25
+ M1L8	2.60 $\pm$ 0.50%	8.62 $\pm$ 0.49%	9.72 $\pm$ 0.14%
+ M4L4	2.57 $\pm$ 0.78%	8.71 $\pm$ 0.58%	9.75 $\pm$ 0.08%
+ M4L8	2.49 $\pm$ 0.66%	8.64 $\pm$ 0.58%	9.79 $\pm$ 0.16%
+ M1L16	2.43 $\pm$ 0.54%	8.73 $\pm$ 0.70%	9.85 $\pm$ 0.16%
+ Gain-only (L8)	2.59 $\pm$ 0.70%	8.61 $\pm$ 0.50%	9.69 $\pm$ 0.17%
+ Gain-only (L16)	2.42 $\pm$ 0.53%	8.66 $\pm$ 0.33%	9.68 $\pm$ 0.18%

NIAH answer NLL by distance bucket ($w=128$, 2,048 tokens)

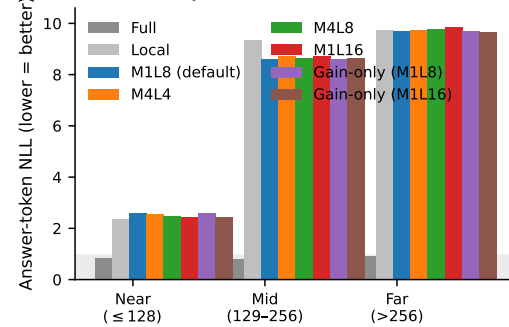


Figure 2: NIAH answer-token NLL by distance bucket (final protocol). Every corrected configuration pays a small precision cost inside the window, gains 7–9% of the local–full gap just beyond it, and recovers essentially nothing in the far bucket — the signature of a smoother.

One niche, however, deserves naming. Below roughly four thousand parameters, generic adapters cannot even be instantiated — a rank-1 LoRA on a single layer’s output projection already costs 4,096 — while 16 gain scalars close 26.4%. In the ultra-low-budget regime, spectral gain calibration has no generic competitor, and future matched-budget adaptation studies should include it as a control.

5.3 What the Gain Is: Smoothing, Not Retrieval

Perplexity can only say that overall language modeling improved; it cannot say whether the model recovered specific distant facts. The needle-in-a-haystack test, placing the answer at controlled distances, paints a subtler picture (Table 4, Figure 2).

Inside the window: a small precision cost. The evidence is right there and local attention was already doing well (2.37 against full attention’s 0.82), yet all six corrected configurations make it slightly worse, by 3.5% to 15.2% of the local–full gap. The

cause is the injection itself: the far branch overlays a smooth average of out-of-window history onto positions where local attention was already sharply focused, diluting that focus. The cost narrows with depth — from -15.2% at M1L8 to -4.3% at M1L16 — which we report as a descriptive fact without a verified mechanism. Especially telling is that the gain-only probes pay almost exactly the same cost as the full structures (-14.3% versus -15.2% at eight layers; -3.5% versus -4.3% at sixteen): the cost travels with the gain channel, not with the kernel.

Just beyond the window: the smoother’s footprint. Local attention is blind at this distance while the exponential kernel’s weight has not yet decayed away, and here all six configurations improve answer NLL by 7% to 9% of the local–full gap. This is the only region where the branch genuinely helps retrieval-style prediction, echoing the sharply position-dependent use of context documented by Liu et al. (2024), and again the mechanism runs through the gain channel — the gain-only probes carry the same signature.

Farther out: nothing. Beyond two windows, no configuration improves by more than 0.8%, and the deepest configuration lands 1.1% worse. One apparent exception: at 1,024 tokens, M1L16 closes $+15.0\%$ of the far-bucket gap (NLL from 10.66 down to 9.21). Yet both numbers remain an order of magnitude above the full-attention baseline, and the effect vanishes entirely at 2,048 tokens (-1.1%) — we read it as improved prior calibration over distant generic text, not evidence retrieval. This mirrors the attribution of the previous section: what the spectral correction recovers is a smooth average of out-of-window information, enough to gently adjust the model’s expectations but not to pick out one specific distant fact. That would require sharp, content-selective attention — exactly what fixed exponential kernels with positive feature maps cannot express by design.

Task metrics supply the other half of the same conclusion. Under official LongBench metrics, all three compared models sit near the floor: full attention averages 0.080, local 0.033, corrected 0.033 (Figure 3). The window’s real cost lands precisely on tasks that demand long-range integration — gov_report’s ROUGE-L falls from 0.137 to 0.003, triviaqa’s F1 from 0.197 to 0.0 — and the failure mode is degenerate repetition (for in-

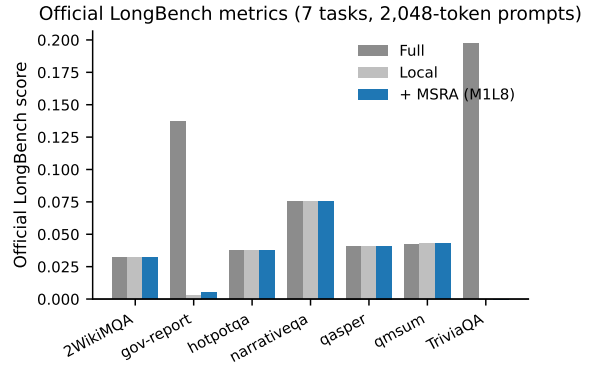


Figure 3: Official LongBench scores per task (F1 / ROUGE-L, greedy decoding, 2,048-token prompts). All three arms sit near the metric floor; the window’s cost concentrates on gov_report and triviaqa, the two tasks demanding genuine long-range integration.

Table 5: Prefill throughput slowdown, each row measured pairwise against a same-session local anchor on the same GPU (local anchors at 512 tokens: 36,956 / 36,991 / 39,532 tok/s). Decode throughput at 2,048 tokens: six paired measurements across two sessions span 0.99–1.10 $\times$ local, median 1.004 — statistically indistinguishable.

Config.	@512	@1,024	@2,048	@4,096
+ M1L8	2.9 $\times$	2.5 $\times$	2.0 $\times$	OOM
+ M1L16	4.8 $\times$	4.1 $\times$	3.0 $\times$	OOM
+ M4L8	8.7 $\times$	6.7 $\times$	5.0 $\times$	OOM

stance “Answer: Answer: . . .”) rather than graceful degradation. On the remaining five tasks all three models generate verbatim-identical floor-level text, where closure is undefined; on the three tasks where local and full differ, the correction closes at most 1.3%. A needle-aware RULER retest — whose full-attention arm passes the positive control at perplexity 1.56, confirming the protocol can detect retrieval when it exists — likewise yields only 3.0% closure. Perplexity gains do not convert into task capability.

5.4 Efficiency

Three patterns emerge from Table 5. The slowdown amortizes with length — local attention’s own cost grows with sequence length while the far branch’s $O(n \log n)$ share shrinks. It scales with the branch’s workload (patched layers times scales: 8, 16, 32 units), reaching 2.9 $\times$, 4.8 $\times$, and 8.7 $\times$ at 512 tokens. And the bottleneck is memory traffic from the FP32 FFT intermediates: peak memory at 2,048 tokens rises from 3.4 GB for local to 13.1 GB for M1L8 and 20.2 GB for M1L16, and

beyond 4,096 tokens every spectral configuration exceeds 24 GB (the local arm itself still runs at 12,457 tok/s).

Decoding tells the opposite story. With the far branch structurally off, the corrected model’s per-token compute is identical to local by construction; six paired measurements across two sessions fall between 0.99 and 1.10 with median 1.004 — statistically indistinguishable from local. The slowdown ratios reproduce across sessions within about 6%, which is the noise floor behind the assertion tolerances.

6 Conclusion

Returning to the opening question — how much of the out-of-window information can a frozen local-attention model recover with a few tens of thousands of retrofitted parameters? — this paper’s complete answer is: a real and stable portion, up to 31.7% of the perplexity gap, but both the composition and the character of that portion deserve a careful look. The gains are dominated by the learned output gains and the grafting depth (16 gain scalars alone close 26.4%, against 31.7% for the full structure); multi-scale kernels and gating contribute genuinely but secondarily. In character, the gain is statistical smoothing of token likelihoods — it improves prediction just beyond the window, slightly sacrifices in-window precision, cannot perform far-distance retrieval, and does not transfer to downstream task metrics. Under budget-matched comparison, generic low-rank adaptation’s 87.4% against 31.7% makes the further point that this gap is mostly generic capacity, for which spectral structure is not a necessary bias.

For practitioners the advice is correspondingly direct. If the goal is to lower a frozen model’s perplexity, choose LoRA — simpler and stronger. If committed to the spectral form, spend the budget on grafting depth rather than kernel diversity. And no such correction should be expected to restore far-distance retrieval. The single exception is the ultra-low-budget niche: below roughly four thousand parameters, generic adapters cannot be instantiated at all, and gain scalars have no competitor there.

For longer-term research, the negative results point forward as well. The far-retrieval failure is structural: fixed kernels with positive feature maps cannot express content selectivity. Retrofitted global pathways that want retrieval capability will likely need addressable memory — compressed

KV stores (Wu et al., 2022) or explicit retrieval (Lewis et al., 2020; Borgeaud et al., 2022) — rather than smooth spectral aggregation. At the same time, the one region where the correction does improve NLL lies just past the window’s edge, exactly where local attention goes blind, suggesting that calibrating representations at the window’s edge is an underappreciated source of gains in low-budget adaptation.

Limitations

Experiments span 1B–1.5B models and training lengths of at most 2,048 tokens, so extrapolation to larger models or longer contexts calls for caution. The retrieval- and task-level analyses (NIAH, LongBench, RULER) were conducted on LLaMA 3.2-1B alone, with GPT2-XL serving only to verify that perplexity gains generalize across architectures. The evaluation uses a single window size ($w=128$); a joint sweep of window and kernel scales is left to future work. LongBench coverage is 7 of 15 English tasks, with the three-way protocol fully matched and no reason to expect the conclusion to change with a larger task set. The multi-scale configuration at sixteen layers was not tested, but the dominance conclusion — LoRA’s 87.4% against the family’s best 31.7% — does not depend on it. The spectral branch’s prefill overhead has seen no kernel-level optimization of the kind developed for IO-aware attention kernels (Dao et al., 2022); the measured slowdowns are upper bounds of an unoptimized implementation, not lower bounds of the method. All implementation-level debugging records, the nine-reversal ledger, and the machine-executed assertion list are collected in the supplementary material.

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